DSA REFERENCE

Two Pointers

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Maintain two positions whose movement discards a provably irrelevant region after every comparison.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

DSA / TWO-POINTERS

DSA

Two Pointers

CORE IDEA

Maintain two positions whose movement discards a provably irrelevant region after every comparison.

Recognition cues

+ sorted input
+ pair or triplet target
+ in-place compaction
+ opposite-end comparison

Implementation sequence

+ name what each pointer represents
+ prove which pointer can move
+ handle equality explicitly
+ decide whether order must be preserved

Invariant

Everything outside the active pointer range has already been classified or cannot improve the answer.

Worked reasoning

In a sorted pair-sum search, a sum below target forces the left pointer right; moving right cannot increase the sum.

Complexity

O(n) after sorting, or O(n log n) including a required sort; often O(1) auxiliary space.

Failure modes

- [ ] moving both pointers without proof
- [ ] ignoring duplicate handling
- [ ] sorting when original indices matter
- [ ] off-by-one loop bounds

Practice ladder

Two Pointers: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Valid Palindrome**	State the cue, invariant, edge case, and proof before implementation.
2	**Two Sum II**	State the cue, invariant, edge case, and proof before implementation.
3	**3Sum**	State the cue, invariant, edge case, and proof before implementation.
4	**Container With Most Water**	State the cue, invariant, edge case, and proof before implementation.
5	**Trapping Rain Water**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS
REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.